

Merrivoss Bioanalysis Inc, a Tarnovell company

OB11 DDI Assessment with CYP3A4 Confirmation

Final study report R1

Study MV-2020-088

Accession M20/BIO/02046

Prepared for Orvessant Biosciences Pty Ltd

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1 Study scope and identification

This final report concerns the OB11 test article set for Orvessant Biosciences Pty Ltd. The study comprised 18 compounds and opened on 04/10/2020. The interim communication was dated 04/20/2020; this R1 report closes interpretation on 04/29/2020.

Table 1. Study scope and DDI benchmarks

Metric	Value
Project	MV-2020-088
Client	Orvessant Biosciences Pty Ltd
Compounds tested	18
Series / chemotype	Orvessant piperidyl benzamide series 11 (Piperidyl benzamide)
Assays	CYP inhibition panel (1A2/2C9/2C19/2D6/3A4) at 10 μ M, CYP3A4 inhibition IC50
Turnaround (calendar days)	19
Price	USD 27,100
CYP1A2 >50% inhibition @10 μ M	4
CYP2C9 >50% inhibition @10 μ M	5
CYP2C19 >50% inhibition @10 μ M	4
CYP2D6 >50% inhibition @10 μ M	3
CYP3A4 >50% inhibition @10 μ M	5
Compounds with any CYP >50%	6
Ketoconazole (3A4) IC50 (μ M)	0.026

The Five-isoform DDI screen and CYP3A4 inhibition curves answer related but distinct questions. Single-concentration inhibition identifies follow-up concerns; concentration-response work defines the CYP3A4 interaction under its probe-substrate conditions. Neither is a measured clinical interaction.

2 Five-isoform DDI screen

The human liver microsomal panel assessed CYP1A2, CYP2C9, CYP2C19, CYP2D6 and CYP3A4 at 10 μ M test article. Values below are percentage inhibition of probe-substrate activity under the screen conditions.

Table 2. Five-isoform inhibition at 10 μ M

Compound	CYP1A2 (%)	CYP2C9 (%)	CYP2C19 (%)	CYP2D6 (%)	CYP3A4 (%)
OB11-08910	10	5	47	4	6
OB11-08918	62	97	100	6	84
OB11-08922	39	67	35	95	53
OB11-08924	1	18	0	27	28
OB11-08928	5	-5	12	8	11
OB11-08942	5	9	26	4	5
OB11-08952	18	-1	9	29	-10
OB11-08954	10	16	33	17	6
OB11-08960	99	93	98	94	85
OB11-08972	100	92	79	33	96
OB11-08994	1	5	-7	18	5
OB11-09014	-0	3	17	5	3
OB11-09024	11	7	10	-2	8
OB11-09026	24	8	18	1	8
OB11-09042	5	9	2	7	3
OB11-09056	66	69	89	49	59
OB11-09062	18	9	3	82	23
OB11-09078	-2	-0	4	5	6

Single concentration 10 μ M, duplicate incubations, mean reported; negative values = no inhibition

Controls (IC₅₀, μ M): furafylline (CYP1A2) 1.7; sulfaphenazole (CYP2C9) 0.29; ticlopidine (CYP2C19) 1.2; quinidine (CYP2D6) 0.061; ketoconazole (CYP3A4) 0.026

The screen uses a common test concentration for comparison, not an IC₅₀ estimate for each isoform. Negative inhibition values remain in the released table; they are not reclassified as a demonstrated activation mechanism.

3 CYP3A4 confirmation table

CYP3A4 inhibition curves used midazolam 1'-hydroxylation as the probe reaction. The reported IC₅₀ is expressed in μ M. Bound entries remain bound statements rather than exact potency estimates.

Table 3. CYP3A4 concentration-response results

Compound	IC ₅₀ (μ M)	Probe substrate
OB11-08910	>50	midazolam
OB11-08918	0.79	midazolam
OB11-08922	9.7	midazolam
OB11-08924	39	midazolam
OB11-08928	>50	midazolam
OB11-08942	>50	midazolam
OB11-08952	>50	midazolam
OB11-08954	>50	midazolam
OB11-08960	1.3	midazolam
OB11-08972	0.83	midazolam
OB11-08994	>50	midazolam
OB11-09014	>50	midazolam
OB11-09024	>50	midazolam
OB11-09026	>50	midazolam
OB11-09042	>50	midazolam
OB11-09056	8.0	midazolam
OB11-09062	38	midazolam
OB11-09078	>50	midazolam

7-point curve, top 50 μ M; ketoconazole control run alongside on every plate

The probe-substrate designation is retained with the values because CYP3A4 inhibition may be substrate-dependent. A curve measured with this reaction is not a general potency value for every CYP3A4 substrate.

4 Matrix and species

Table 4. Matrix and readout definitions

Assessment	Biological matrix	Species	Readout
Five-isoform DDI screen	Liver microsomes	Human	% inhibition at 10 μ M
CYP3A4 inhibition curves	Liver microsomal probe-reaction context	Human	IC50 in μ M; midazolam 1'-hydroxylation

No rat inhibition table or whole-cell induction result is supplied. These are enzyme-inhibition observations in a human in vitro context. They do not establish the concentration of test article at the enzyme in a patient.

5 Accession-lot and parent identity

The controlling accession is M20/BIO/02046. Complete OB11 identifiers connect the inhibition screen, the confirmation table and the companion assay worksheet. The submitted parent is the analyte designation used throughout.

The comparison assumes consistent test article identity between screen and curve work. A later chemistry change must retain a predecessor link while receiving its own compound and accession identity. The present results cannot be transferred to a new cap or branch without measurement.

6 Screen procedure and interpretation

The panel measures reduction of probe-substrate activity relative to the assay reference at 10 μ M test article. Because the test concentration is fixed, the screen is useful for prioritization but does not supply a full concentration-response relationship.

The five isoforms are retained separately. A concern in one column is not proof of equivalent inhibition in the others. The CYP3A4 confirmation is a targeted follow-up and does not replace the remaining screen columns.

7 Concentration-response procedure

The CYP3A4 follow-up assesses activity across concentrations and reports the resulting IC₅₀ or limit statement. The probe is midazolam, and the reaction is 1'-hydroxylation. No alternative probe curve is supplied.

An IC₅₀ describes inhibition within the stated assay context. No K_i, time-dependent inactivation constant or mechanistic inhibition class is derived here. Those quantities require experiments beyond this screen-and-confirmation design.

8 Reference-response benchmarks

The released reference-response values are shown with their historical comparators. The comparison supports a reading of assay behavior without introducing a numerical acceptance band absent from the supplied record.

Table 5. Reference response and historical comparison

Isoform	Current reference response (µM)	Historical comparator (µM)
CYP1A2	1.7	1.8
CYP2C9	0.29	0.35
CYP2C19	1.2	1.2
CYP2D6	0.061	0.060
CYP3A4	0.026	0.030

Current and historical values are not test article results. Their proximity or difference cannot, by itself, establish that every underlying analytical sample passed a numerical criterion.

9 Screen summary by isoform

At the 50% screen flag, the numbers of compounds above the flag are 4 for CYP1A2, 5 for CYP2C9, 4 for CYP2C19, 3 for CYP2D6 and 5 for CYP3A4.

The number above the flag in any isoform is 6. Counts across columns overlap and must not be summed as a count of distinct compounds. The benchmark identifies CYP2C9 and CYP3A4 as the most flagged isoform; that designation is not a declaration that other isoforms are free of concern.

10 CYP3A4 confirmation summary

The concentration-response release includes 18 compounds. The number with IC₅₀ below 1 µM is 2; the number below 10 µM is 5. These are nested categories and must not be added.

The screen percentage and curve IC₅₀ have different units and meanings. Their agreement is assessed by direction and relative ordering, not by treating a percentage as a concentration or forcing a numeric conversion between them.

11 Analytical measurement responsibilities

Javier Quintana is the mass-spectrometry lead. Analyte cards and the assay worksheet retain matrix-associated analytical details and parent assignment. The report provides released inhibition results rather than reproducing individual analytical acquisitions.

Interference with probe-product measurement can resemble inhibition if analytical response is not kept separate from enzyme activity. The final narrative therefore retains the matrix context and does not assign mechanism solely from a reduction in measured activity.

12 Analytical limits

Numerical quantitation limits, calibration ranges and raw response values are not supplied with the final tables. The report does not create them. The result release supports comparison of inhibition, not reconstruction of every probe-product measurement.

A greater-than IC50 entry indicates that the reported boundary was not crossed within the released range. It is neither zero inhibition nor an exact IC50 at the printed limit. Negative screen values likewise remain part of the signed data presentation.

13 Matrix acceptance and exceptions

Helen Marchwood is responsible for bioanalytical quality. The companion assay worksheet contains matrix acceptance checks. Reference-response benchmarks are retained above as the context supplied with this final result record.

No named sample-level exception, repeat history or numerical acceptance interval is supplied. None is inferred. The report does not equate a favorable test article profile with acceptance of all analytical samples, nor add an exclusion list that is absent from the record.

14 Comparison of screen and follow-up

For the OB11 cap decision, the CYP3A4 follow-up confirmed the interaction concern in the lipophilic cap variants. Concentration-response work supplied a more specific basis than the single-concentration screen alone.

Confirmation remains probe-specific and CYP3A4-specific. It does not supply concentration-response results for the other isoforms. The complete panel should remain available when the client considers later chemistry choices.

15 Chemistry decision basis

The CYP3A4 follow-up confirmed the liability in the lipophilic cap variants and Orvessant switched to the alternate cap. The cap choice is the client's recorded response to the confirmed in vitro concern.

The alternate cap is a next chemistry direction. No result for newly prepared alternate-cap material is supplied, and switching the design does not by itself prove that the next test article set will meet an exposure gate.

16 Exposure context and DDI boundaries

The study contains no circulating exposure measurement, unbound clinical concentration or administered-dose record. An in vitro IC50 cannot be converted into clinical interaction risk without that additional context.

No induction, time-dependent inhibition or transporter assessment is reported. The term DDI screen is the house name for this inhibition panel; it must not be expanded into a claim that all mechanisms of interaction have been evaluated.

17 Handoff to later work

The alternate-cap handoff should preserve a predecessor link to the lipophilic cap variants. New test articles require their own accession and inhibition records so that the chemistry response can be assessed against this baseline.

Later work should keep the probe reaction and test concentration visible when comparing results. A different assay context can answer a useful question but should not be treated as identical evidence without qualification.

18 Final conclusion

The CYP3A4 follow-up confirmed the liability in the lipophilic cap variants and Orvessant switched to the alternate cap. Megan Larkin records this as the final study outcome for Orvessant Biosciences Pty Ltd under MV-2020-088.

The decision is supported within the scope of the released human microsomal inhibition data. Matrix definitions, probe specificity, limit handling and analytical qualifications remain part of any extract used for the exposure gate.

19 Report control and assumptions

This final R1 report is dated 04/29/2020. The companion MV-2020-088_AssayWorksheet_042920_v03.xlsx holds accession-lot mapping and matrix acceptance. The 04/20/2020 interim communication is superseded for closeout by this final interpretation.

- Preserve complete compound and accession identifiers.
- Do not sum overlapping isoform flag counts.
- Keep nested curve counts distinct.
- Retain bound signs and negative screen values.
- Keep the midazolam probe reaction with CYP3A4 IC50 results.
- Do not extend this inhibition release to clinical risk or untested chemistry.

Name	Role	Signature	Date
Megan Larkin	DMPK Study Director	<u>M Larkin</u>	04/29/2020
Javier Quintana	Mass-Spectrometry Manager	<u>J Quintana</u>	04/29/2020
Helen Marchwood	Bioanalytical Quality Manager	<u>H Marchwood</u>	04/29/2020